RapidFuzz

Fuzzy string comparisons beside your data.

Install & load

```
SET allow_extension_repositories = 'allowed';
CREATE EXTENSION REPOSITORY IF NOT EXISTS query_farm WITH PREFIX 'https://duckdb-extensions.query.farm';
INSTALL rapidfuzz FROM query_farm;  LOAD rapidfuzz FROM query_farm;
```

01 At a glance

Score candidate pairs using edit-distance and token-based matching, compared by Unicode character. Normalize text first, select a score that matches your data, then rank, threshold, or pick the best match from a list.

Normalize text → Score candidates → Rank matches

02 Core SQL surface

`rapidfuzz_ratio(a, b)`

General string similarity on a 0–100 scale.

`rapidfuzz_partial_ratio(a, b)`

Find a shorter string within a longer string.

`rapidfuzz_token_sort_ratio(a, b)`

Compare after sorting words; tolerates word-order changes.

`rapidfuzz_token_set_ratio(a, b)`

Compare token sets; discounts repeated words.

`rapidfuzz_levenshtein_distance(a, b)`

Edits as BIGINT; costs and score_cutoff are named options.

`rapidfuzz_osa_distance(a, b)`

Edit distance with adjacent transpositions; smaller is closer.

`rapidfuzz_jaro_winkler_normalized_similarity`

Normalized similarity from 0 to 1; larger is closer.

`rapidfuzz_extract_one(query, choices)`

Best match from a list: choice, score and choice_index.

03 Keep in mind

Score scales differ

Ratio functions use 0–100 and normalized similarities 0–1. Counting distances return BIGINT and point the other way: zero is an exact match. Characters, not bytes, are compared, so café to cafe is one edit. A score is not a match probability.

04 Reordered words

```
SELECT rapidfuzz_token_sort_ratio(
  'world hello', 'hello world'
) AS score;
-- 100
```

Token sorting handles word order without changing the score scale.

05 Find the closest candidate

```
SELECT name,
  rapidfuzz_ratio('query farm', lower(name))
    AS score
FROM (VALUES ('Query Farm'), ('Query Firm'),
             ('Quiet Barn')) AS t(name)
ORDER BY score DESC
LIMIT 2;
```

This small fixture is self-contained. Apply the same pattern to blocked pairs.

06 Measure a transposition

```
SELECT rapidfuzz_osa_distance(
  'form', 'from'
) AS edits;
-- 1
```

OSA counts one adjacent transposition as one edit.

07 Pick the best from a list

```
SELECT rapidfuzz_extract_one('acme corp',
  ['acme co', 'acme corp.', 'apex corp'],
  scorer := 'ratio'
) AS best;
-- {choice: acme corp., score: 94.7…, choice_index: 2}
```

No join needed; rapidfuzz_extract returns the top_k matches as a list.

08 In practice

Reduce candidate pairs first

A cross join compares every pair. Block on useful fields before scoring large tables. Explicitly lower-case, trim, or standardize punctuation when your matching rules require it.

Full docs & examples ↗ Source ↗